

# ECO101

Introduction to Microeconomics

Yamim Rashad Nawaz

BRAC University

Summer 2026



Credit: Cengage 2014

## *Week 3 – Demand & Supply*

- *What factors affect buyers' demand for goods?*
- *What factors affect sellers' supply of goods?*
- *How do supply and demand determine the price of a good and the quantity sold?*
- *How do changes in the factors that affect demand or supply affect the market price and quantity of a good?*
- *How do markets allocate resources?*

# *Markets and Competition*

A **market** is a group of buyers and sellers of a particular product.

A **competitive market** is one with many buyers and sellers, each has a negligible effect on price.

In a **perfectly competitive** market:

- All goods exactly the same
- Buyers & sellers so numerous that no one can affect market price — each is a “**price taker**”

In this lecture, we assume markets are perfectly competitive.

# *Demand*

The **quantity demanded** of any good is the amount of the good that buyers are willing and able to purchase.

**Law of demand:** the claim that the quantity demanded of a good falls when the price of the good rises, other things equal

# The Demand Schedule

- **Demand schedule:**  
a table that shows the relationship between the price of a good and the quantity demanded
- Example:  
Ash's demand for lattes.
- Notice that Ash's preferences obey the law of demand.

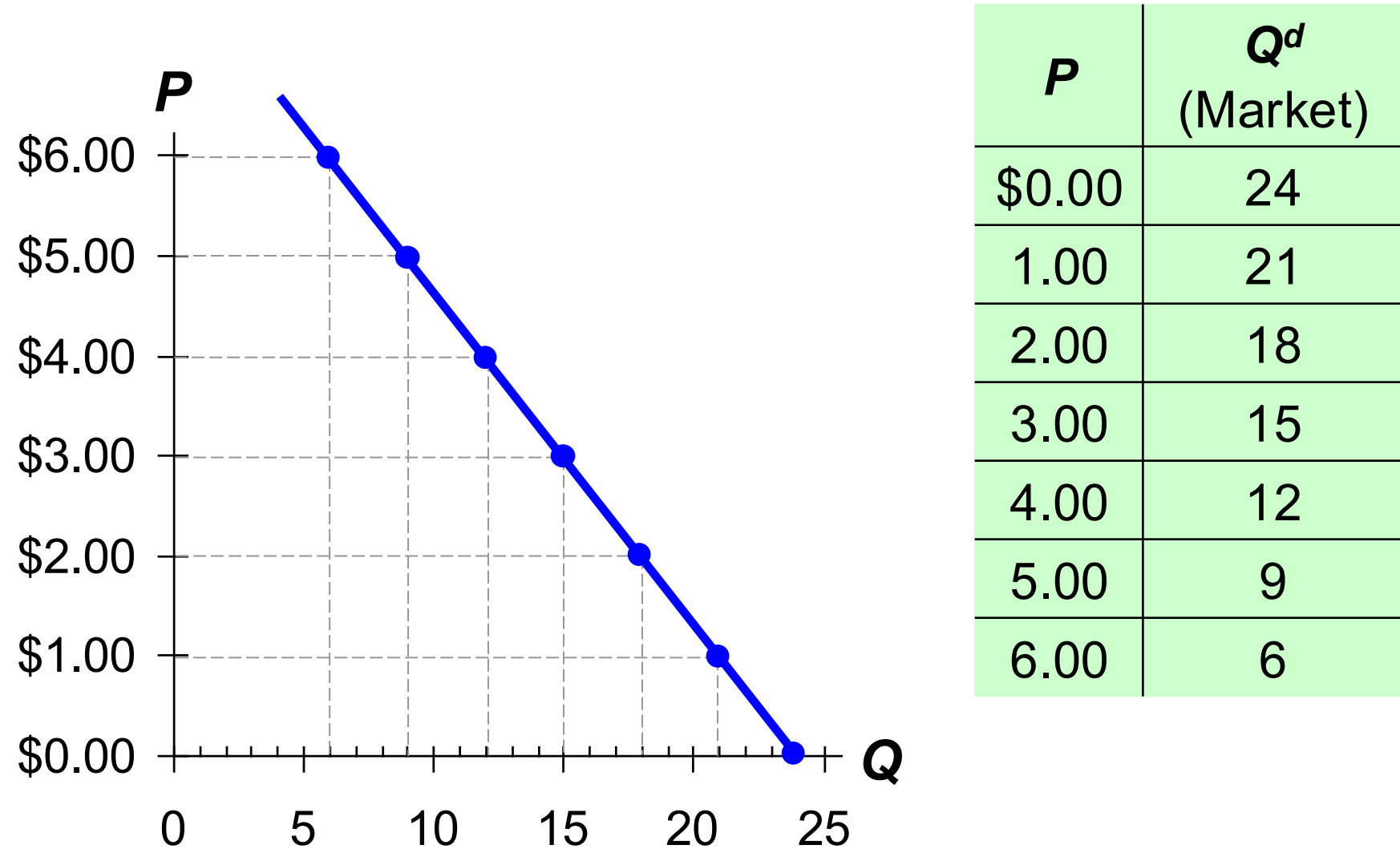
| Price of lattes | Quantity of lattes demanded |
|-----------------|-----------------------------|
| \$0.00          | 16                          |
| 1.00            | 14                          |
| 2.00            | 12                          |
| 3.00            | 10                          |
| 4.00            | 8                           |
| 5.00            | 6                           |
| 6.00            | 4                           |

# Market Demand versus Individual Demand

- The quantity demanded in the market is the sum of the quantities demanded by all buyers at each price.
- Suppose Ash and Gary are the only two buyers in the Latte market. ( $Q^d$  = quantity demanded)

| Price  | Ash's $Q^d$ |   | Gary's $Q^d$ |   | Market $Q^d$ |
|--------|-------------|---|--------------|---|--------------|
| \$0.00 | 16          | + | 8            | = | 24           |
| 1.00   | 14          | + | 7            | = | 21           |
| 2.00   | 12          | + | 6            | = | 18           |
| 3.00   | 10          | + | 5            | = | 15           |
| 4.00   | 8           | + | 4            | = | 12           |
| 5.00   | 6           | + | 3            | = | 9            |
| 6.00   | 4           | + | 2            | = | 6            |

# The Market Demand Curve for Lattes



| $P$    | $Q^d$<br>(Market) |
|--------|-------------------|
| \$0.00 | 24                |
| 1.00   | 21                |
| 2.00   | 18                |
| 3.00   | 15                |
| 4.00   | 12                |
| 5.00   | 9                 |
| 6.00   | 6                 |

## Demand Curve Shifters

The demand curve shows how **price affects quantity demanded**, *other things being equal*.

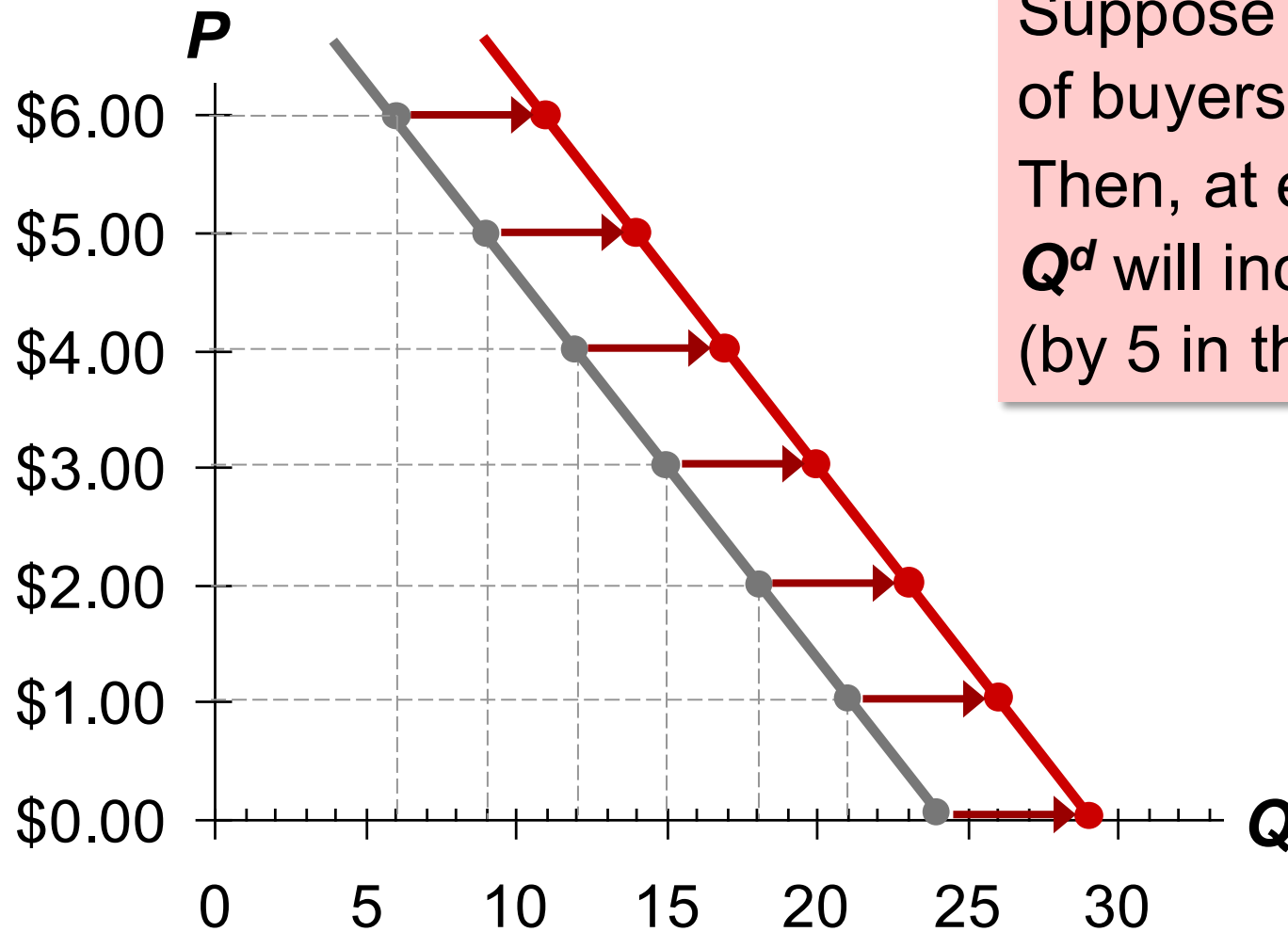
These “other things” are **non-price determinants of demand** (i.e., things that determine buyers’ demand for a good, other than the good’s price).

Changes in them shift the *D* curve.

## Demand Curve Shifters: # of Buyers

**Increase in # of buyers** increases quantity demanded at each price, shifts *D* curve to the right.

## Demand Curve Shifters: # of Buyers



Suppose the number of buyers increases. Then, at each  $P$ ,  $Q^d$  will increase (by 5 in this example).

## Demand Curve

### Shifters:

#### Income

Demand for a **normal good** is positively related to income.

Increase in income causes increase in quantity demanded at each price, shifts ***D*** curve to the right.

(Demand for an **inferior good** is negatively related to income. An increase in income shifts ***D*** curves for inferior goods to the left.)

# Demand Curve Shifters:

## Prices of Related Goods

Two goods are **substitutes**:

if an increase in the price of one causes an increase in demand for the other.

Example: pizza and hamburgers.

An increase in the price of pizza increases demand for hamburgers, shifting hamburger demand curve to the right.

Other examples: Coke and Pepsi, laptops and desktop computers, Spotify Premium and Apple Music

## Demand Curve

### Shifters:

### Prices of Related Goods

Two goods are **complements**:  
if an increase in the price of one  
causes a fall in demand for the other.

Example: computers and software. If price of computers rises, people buy fewer computers, and therefore, less software. Software demand curve shifts left.

Other examples: college tuition and textbooks,  
bagels and cream cheese, eggs and bacon

# Demand Curve Shifters:

## Tastes

Anything that causes a shift in tastes *toward* a good will increase demand for that good and shift its *D* curve to the right.

Example:

Online classrooms and meetings have become more prevalent since covid thus increasing the demand for platforms that offer these services.

# **Demand Curve**

## **Shifters:**

### **Expectations**

**Expectations affect consumers' buying decisions.**

#### **Examples:**

If people expect their incomes to rise, their demand for meals at expensive restaurants may increase now.

If the economy sours and people worry about their future job security, demand for new autos may fall now.

## Summary: Variables That Influence Buyers

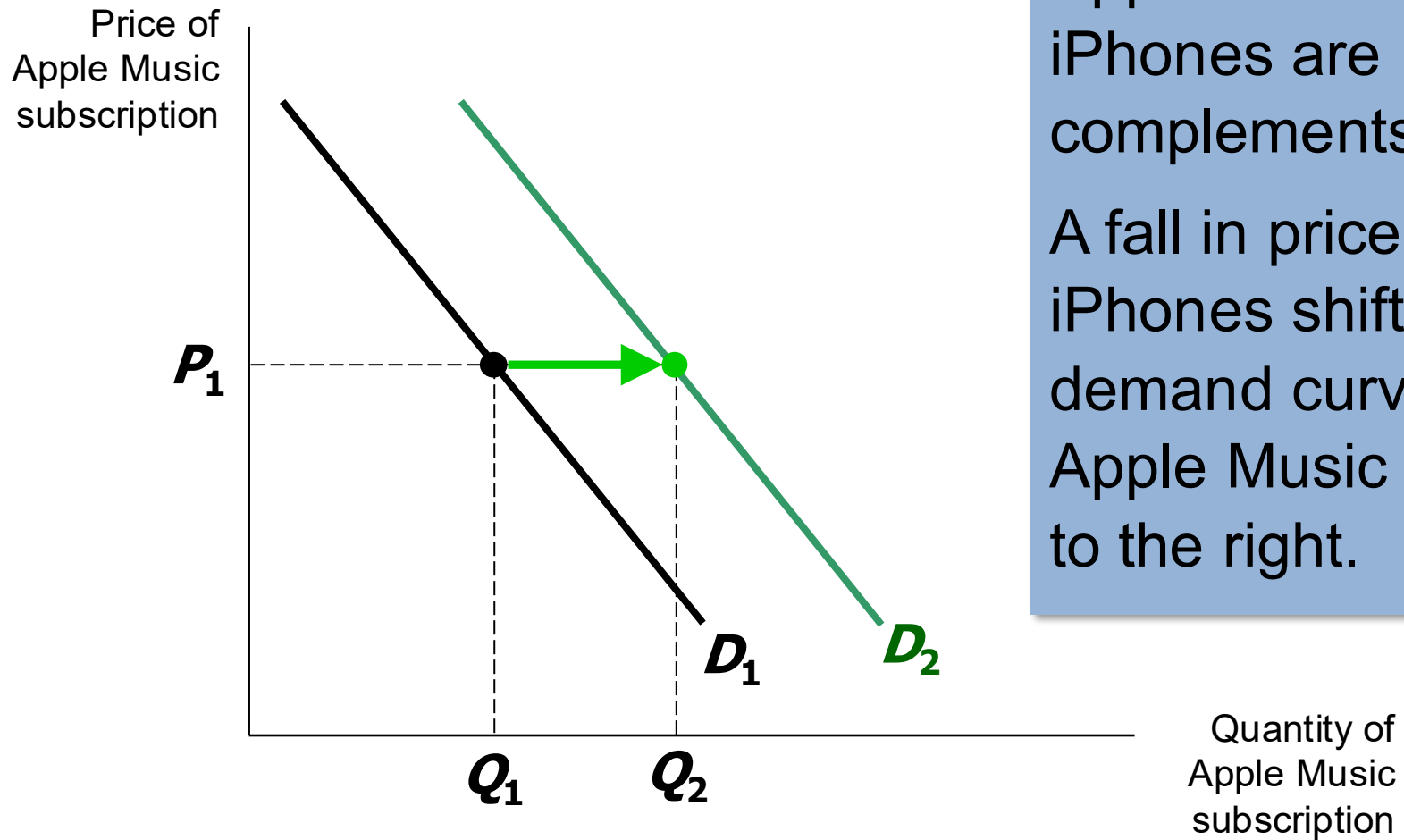
| Variable               | A change in this variable...                         |
|------------------------|------------------------------------------------------|
| Price                  | ...causes a movement along the <b><i>D</i></b> curve |
| # of buyers            | ...shifts the <b><i>D</i></b> curve                  |
| Income                 | ...shifts the <b><i>D</i></b> curve                  |
| Price of related goods | ...shifts the <b><i>D</i></b> curve                  |
| Tastes                 | ...shifts the <b><i>D</i></b> curve                  |
| Expectations           | ...shifts the <b><i>D</i></b> curve                  |

## Food for thought

**Demand curve for Apple Music.**

- A.** The price of iPhones falls
- B.** The price of Apple Music subscription falls
- C.** The price of Spotify Premium falls

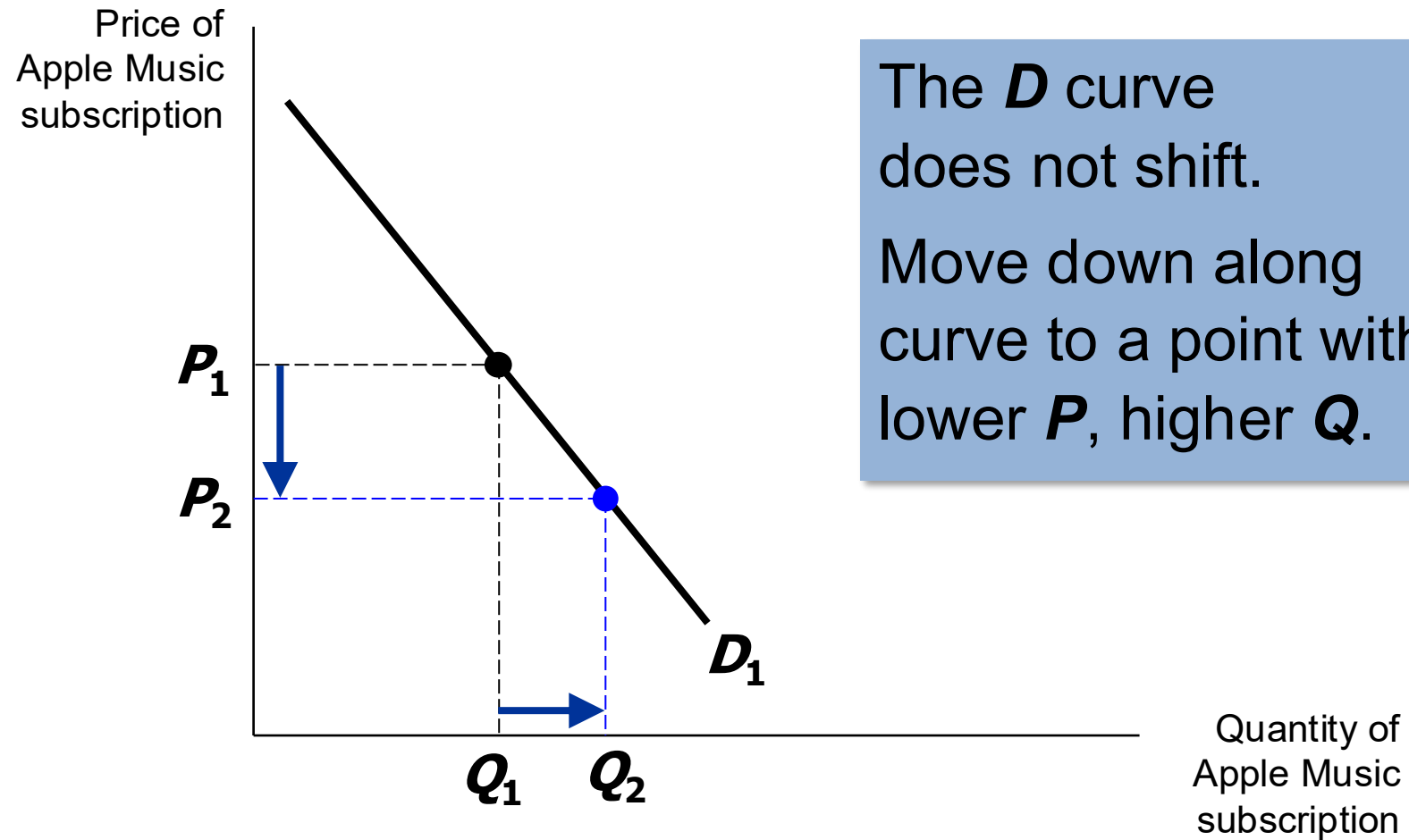
## A. Price of iPhones falls



Apple Music and iPhones are complements.

A fall in price of iPhones shifts the demand curve for Apple Music to the right.

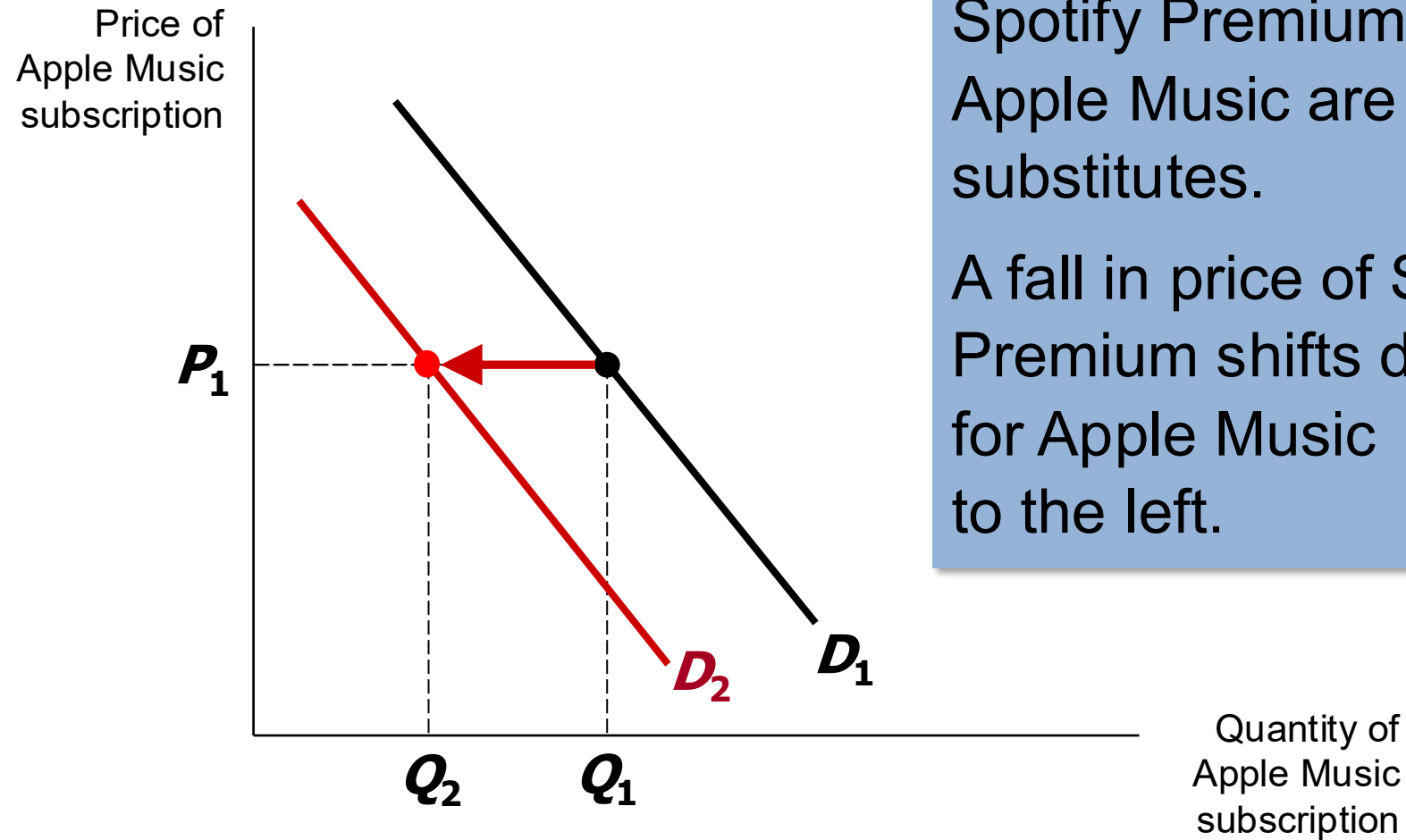
## B. Price of Apple Music subscription falls



The  $D$  curve does not shift.

Move down along curve to a point with lower  $P$ , higher  $Q$ .

## C. Price of Spotify Premium falls



Spotify Premium and Apple Music are substitutes.

A fall in price of Spotify Premium shifts demand for Apple Music to the left.

# *Supply*

The **quantity supplied** of any good is the amount that sellers are willing and able to sell.

**Law of supply:** the claim that the quantity supplied of a good rises when the price of the good rises, other things equal

# The Supply Schedule

- **Supply schedule:**  
A table that shows the relationship between the price of a good and the quantity supplied.
- Example:  
Starbucks' supply of lattes.
- Notice that Starbucks' supply schedule obeys the law of supply.

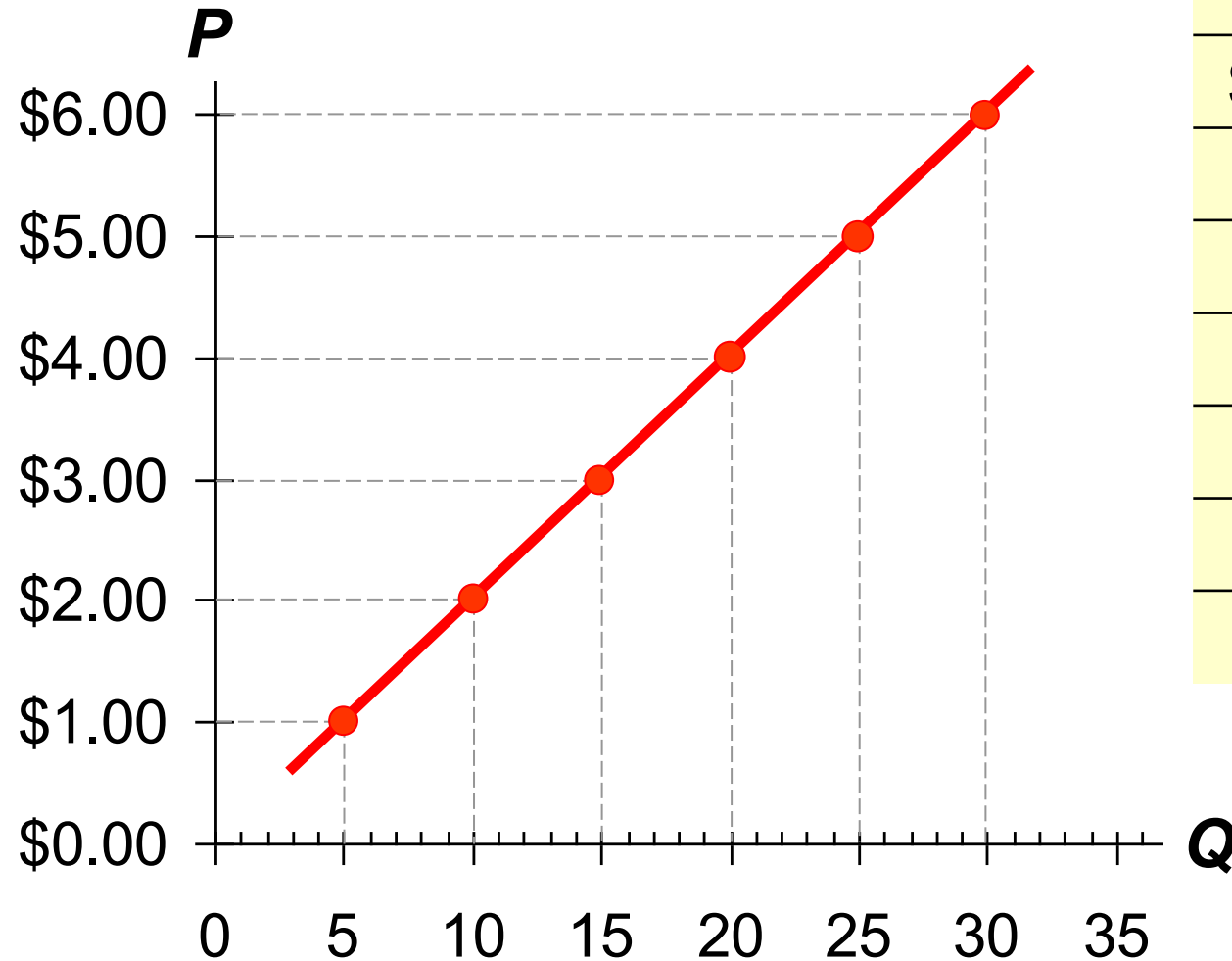
| Price of lattes | Quantity of lattes supplied |
|-----------------|-----------------------------|
| \$0.00          | 0                           |
| 1.00            | 3                           |
| 2.00            | 6                           |
| 3.00            | 9                           |
| 4.00            | 12                          |
| 5.00            | 15                          |
| 6.00            | 18                          |

# Market Supply versus Individual Supply

- The quantity supplied in the market is the sum of the quantities supplied by all sellers at each price.
- Suppose Starbucks and Jitters are the only two sellers in this market. ( $Q^s$  = quantity supplied)

| Price  | Starbucks |   | Jitters |   | Market $Q^s$ |
|--------|-----------|---|---------|---|--------------|
| \$0.00 | 0         | + | 0       | = | 0            |
| 1.00   | 3         | + | 2       | = | 5            |
| 2.00   | 6         | + | 4       | = | 10           |
| 3.00   | 9         | + | 6       | = | 15           |
| 4.00   | 12        | + | 8       | = | 20           |
| 5.00   | 15        | + | 10      | = | 25           |
| 6.00   | 18        | + | 12      | = | 30           |

# The Market Supply Curve



| $P$    | $Q^s$<br>(Market) |
|--------|-------------------|
| \$0.00 | 0                 |
| 1.00   | 5                 |
| 2.00   | 10                |
| 3.00   | 15                |
| 4.00   | 20                |
| 5.00   | 25                |
| 6.00   | 30                |

## **Supply Curve Shifters**

**The supply curve shows how price affects quantity supplied, other things being equal.**

**These “other things” are non-price determinants of supply.**

**Changes in them shift the S curve...**

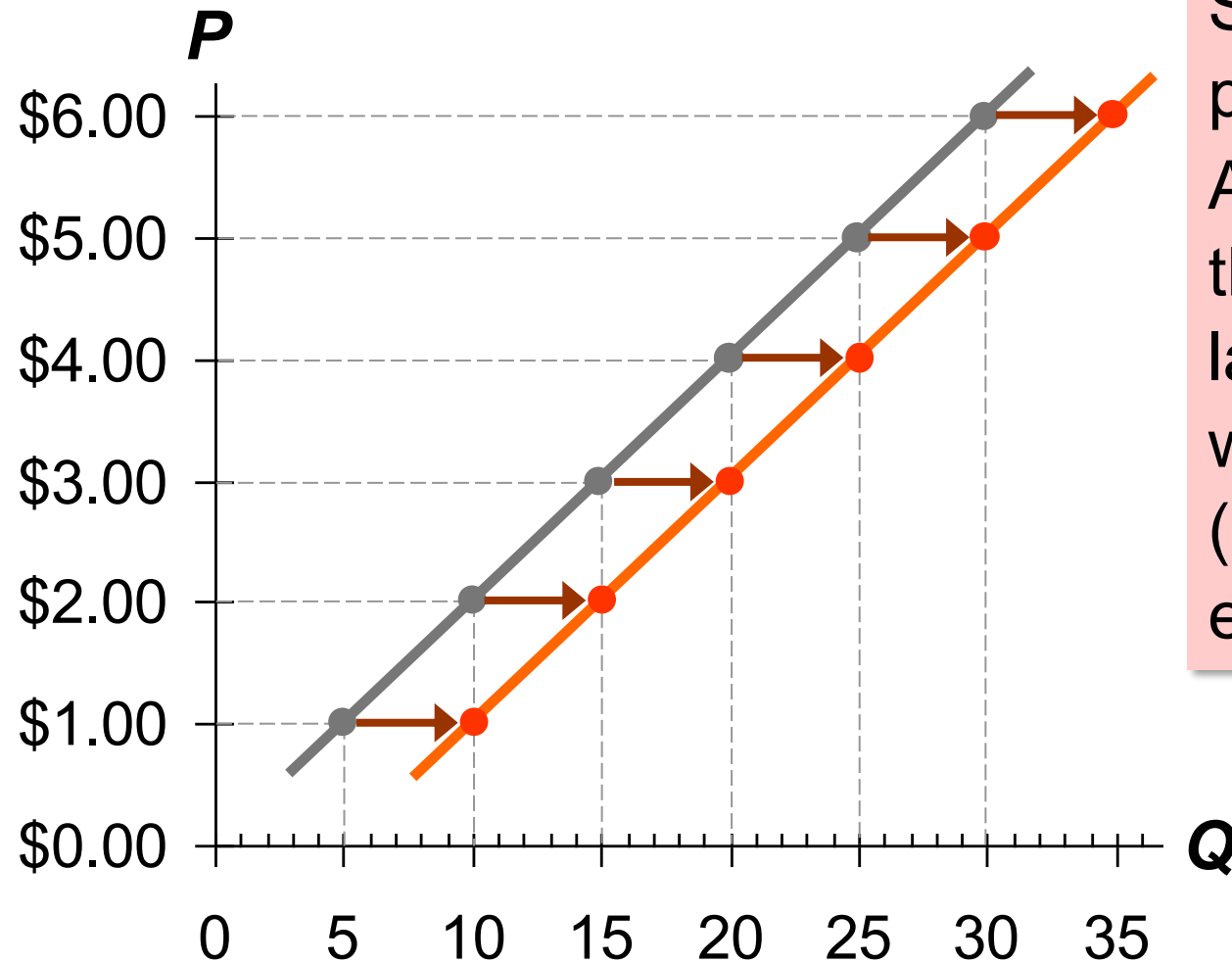
## Supply Curve Shifters: Input Prices

Examples of input prices:  
*wages, prices of raw materials.*

A **fall in input prices makes production more profitable at each output price**, so firms supply a larger quantity at each price, and the S curve shifts to the right.

A rise in input prices does the opposite.

# Supply Curve Shifters: Input Prices



Suppose the price of milk falls. At each price, the quantity of lattes supplied will increase (by 5 in this example).

# Supply Curve

Shifters:

**Technology**

Technology determines how much inputs are required to produce a unit of output.

A cost-saving technological improvement has the same effect as a fall in input prices, shifts *S* curve to the right.

# Supply Curve Shifters: # of Sellers

An increase in the number of sellers increases the quantity supplied at each price, shifts *S* curve to the right.

# Supply Curve Shifters: Expectations

## **Example:**

Events in the Middle East lead to expectations of higher oil prices.

In response, owners of Texas oilfields reduce supply now, save some inventory to sell later at the higher price.

**S** curve shifts left.

**In general, sellers may adjust supply\* when their expectations of future prices change. (\*If good not perishable)**

## Summary: Variables that Influence Sellers

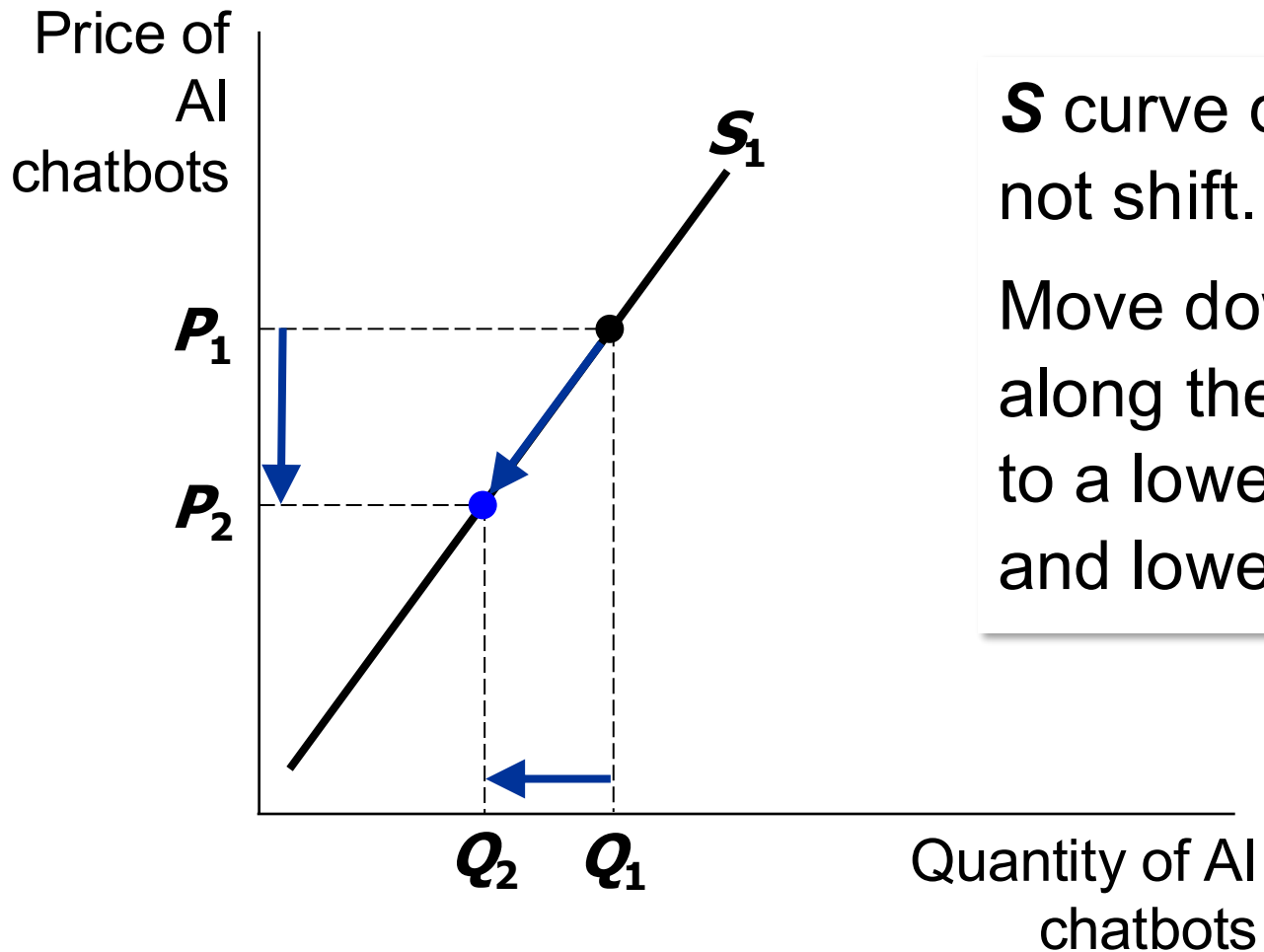
| Variable     | A change in this variable...                  |
|--------------|-----------------------------------------------|
| Price        | ...causes a movement along the <b>S</b> curve |
| Input Prices | ...shifts the <b>S</b> curve                  |
| Technology   | ...shifts the <b>S</b> curve                  |
| # of Sellers | ...shifts the <b>S</b> curve                  |
| Expectations | ...shifts the <b>S</b> curve                  |

## Food for thought

**Draw a supply curve for AI chatbots.**

- A.** Firms cut the price of the software.
- B.** A technological advance allows the software to be produced at lower cost.
- C.** Professional AI users raise the price of the services they provide.

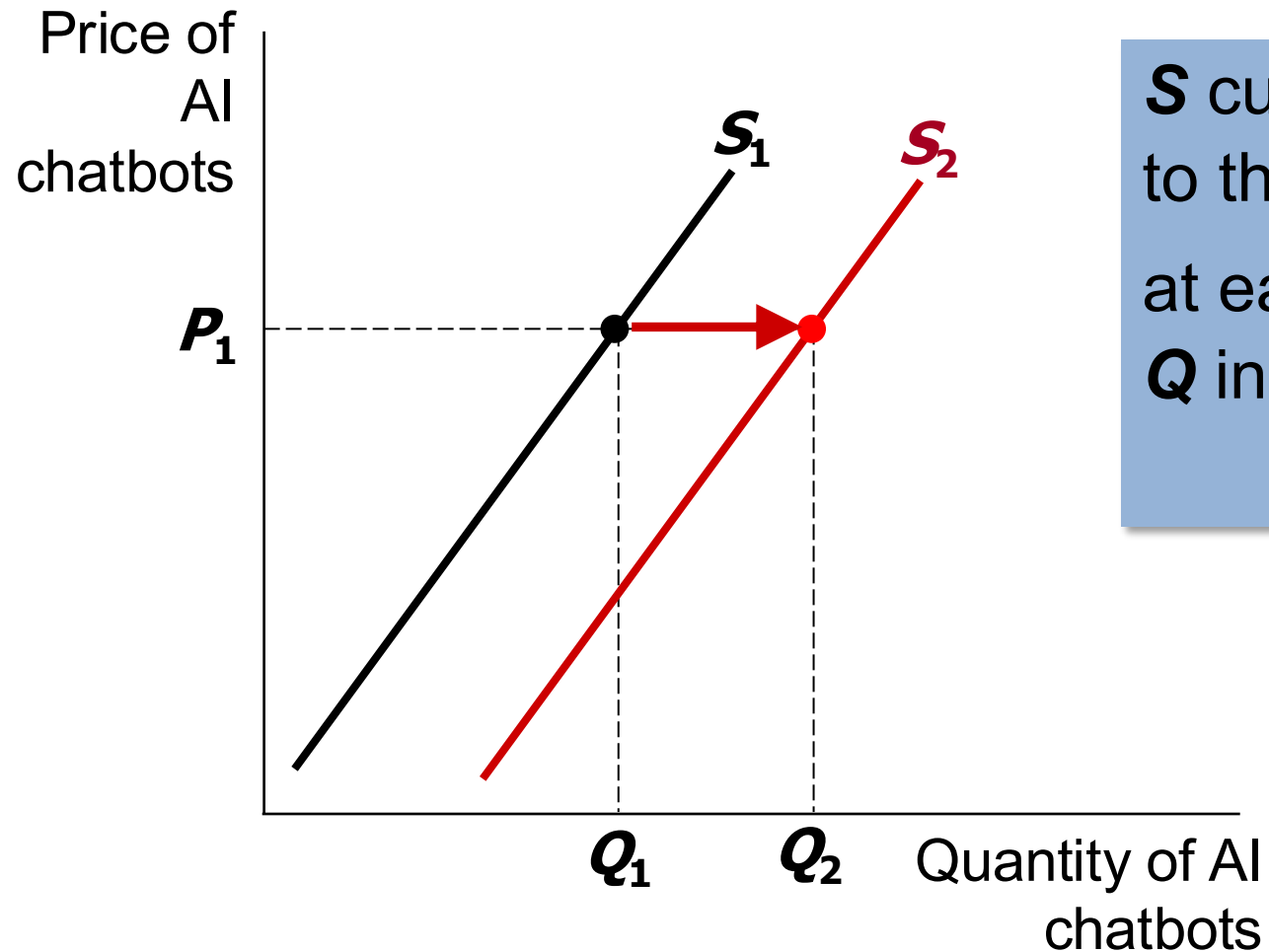
## A. Fall in price of AI chatbots



**S** curve does not shift.

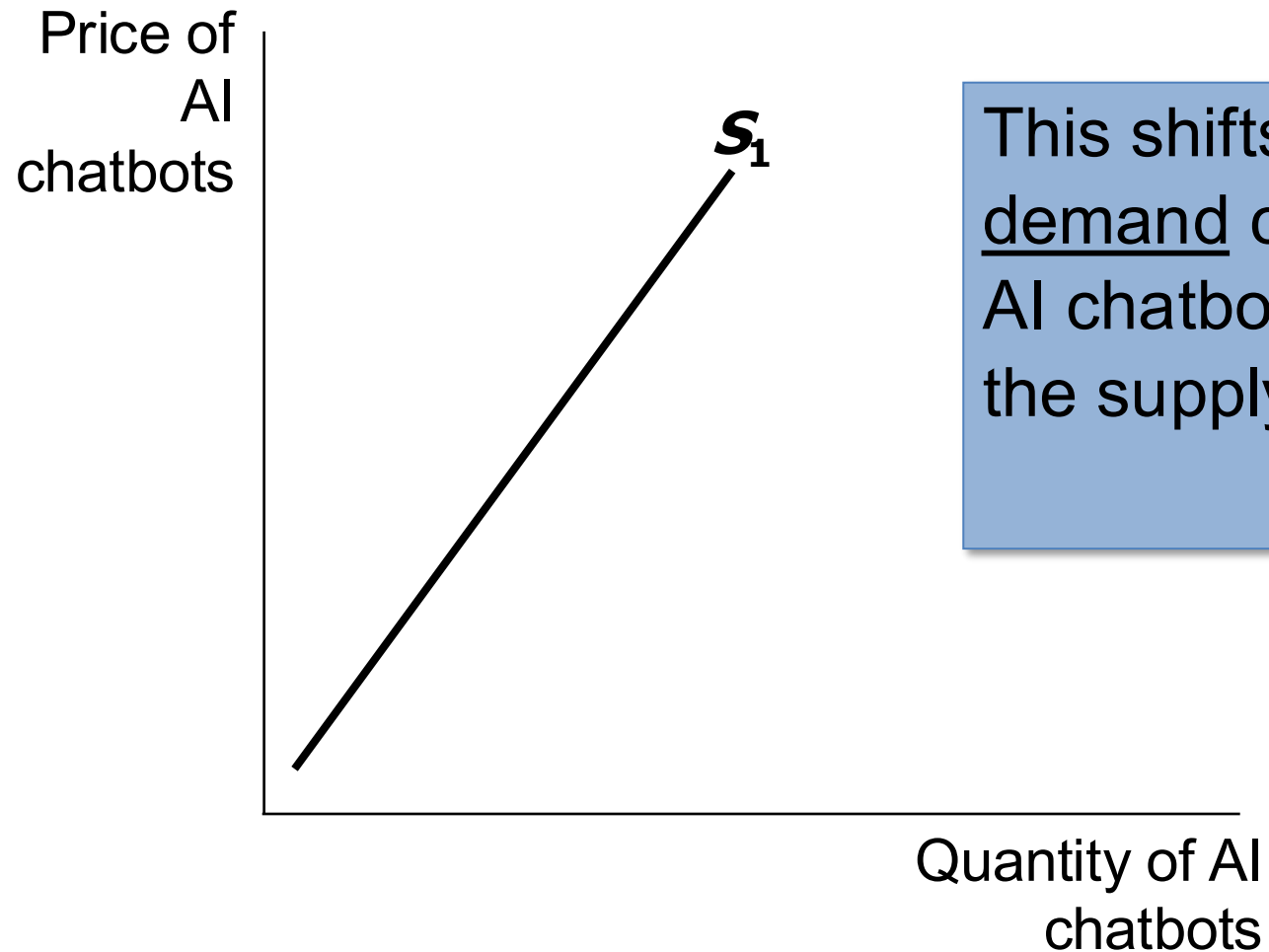
Move down along the curve to a lower **P** and lower **Q**.

## B. Fall in cost of producing the software



**S** curve shifts to the right: at each price,  $Q$  increases.

## C. Professional AI users raise their price

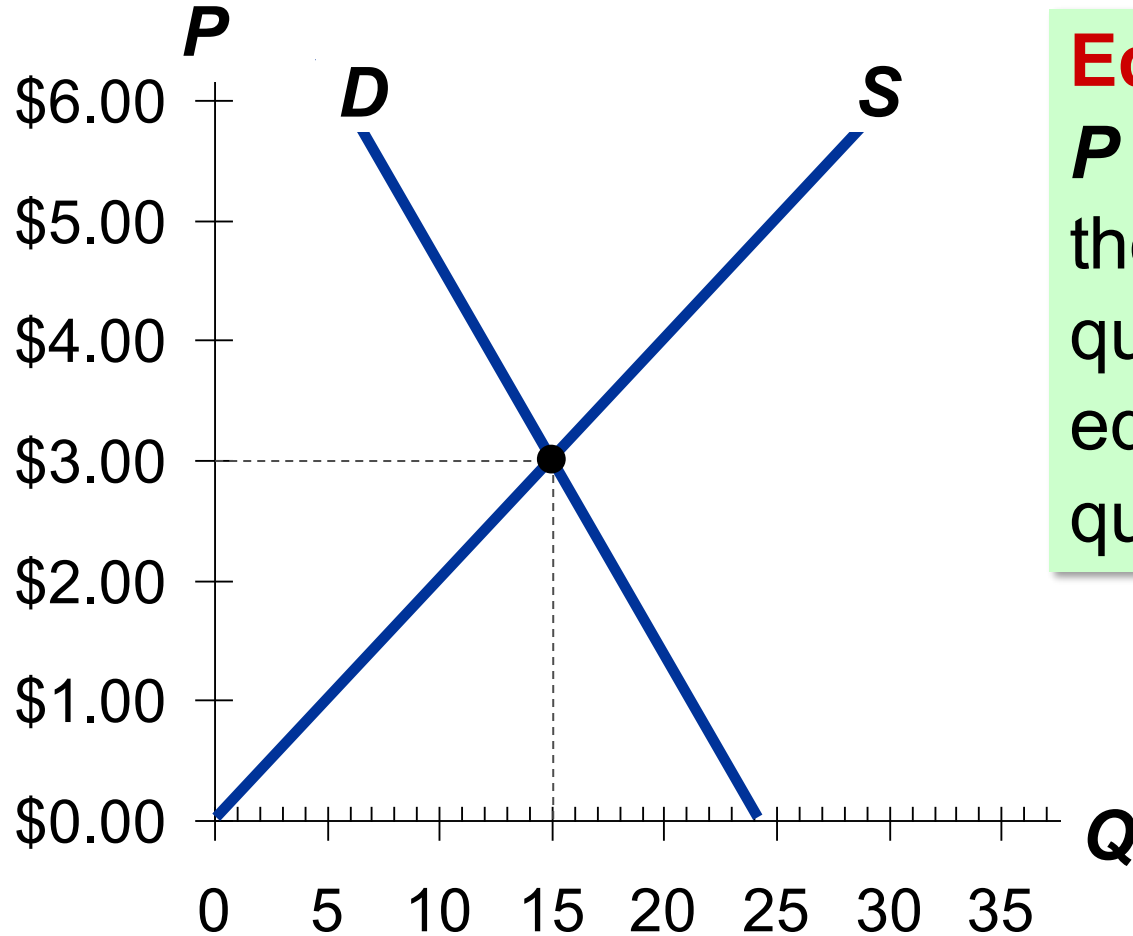


This shifts the demand curve for AI chatbots, not the supply curve.

# ***Market Equilibrium***

Putting Demand and Supply together

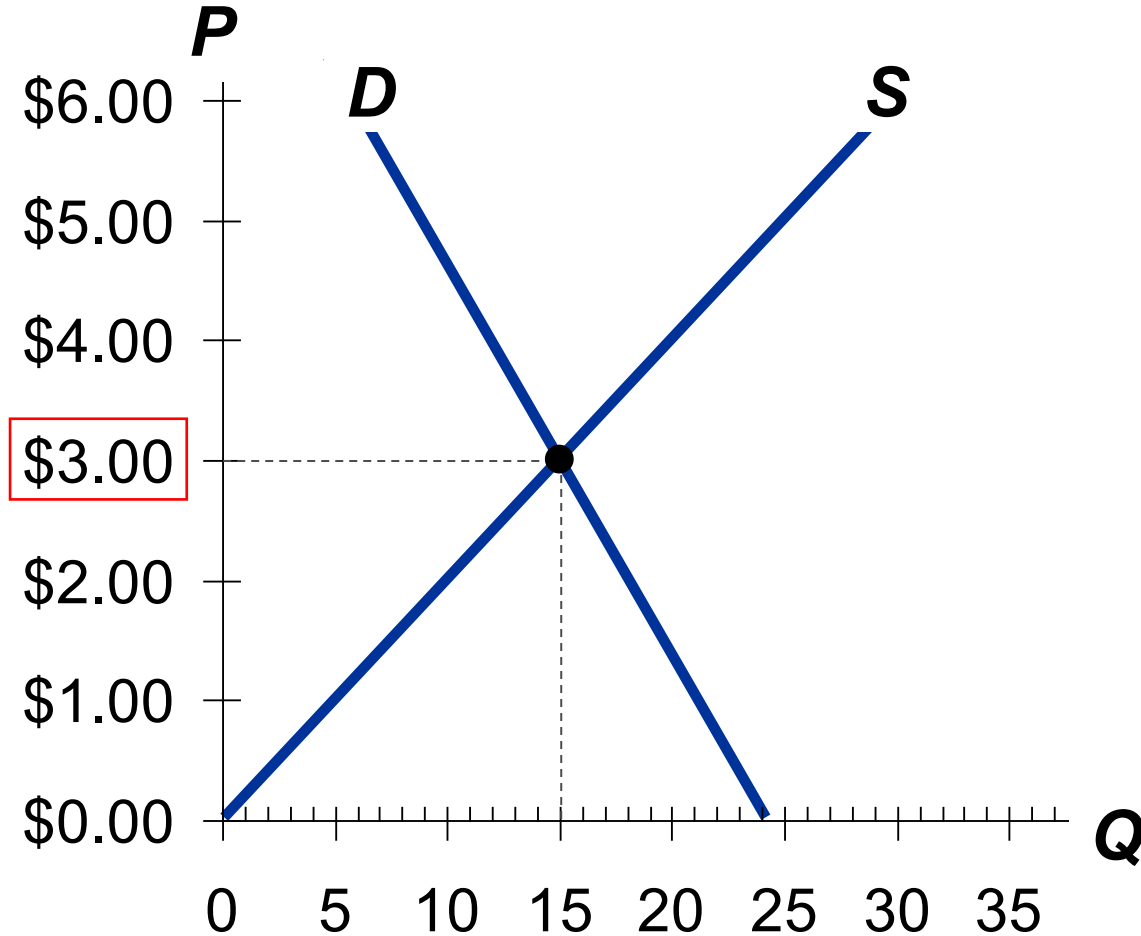
# Supply and Demand Together



**Equilibrium:**  
 $P$  has reached  
the level where  
quantity supplied  
equals  
quantity demanded

## Equilibrium price:

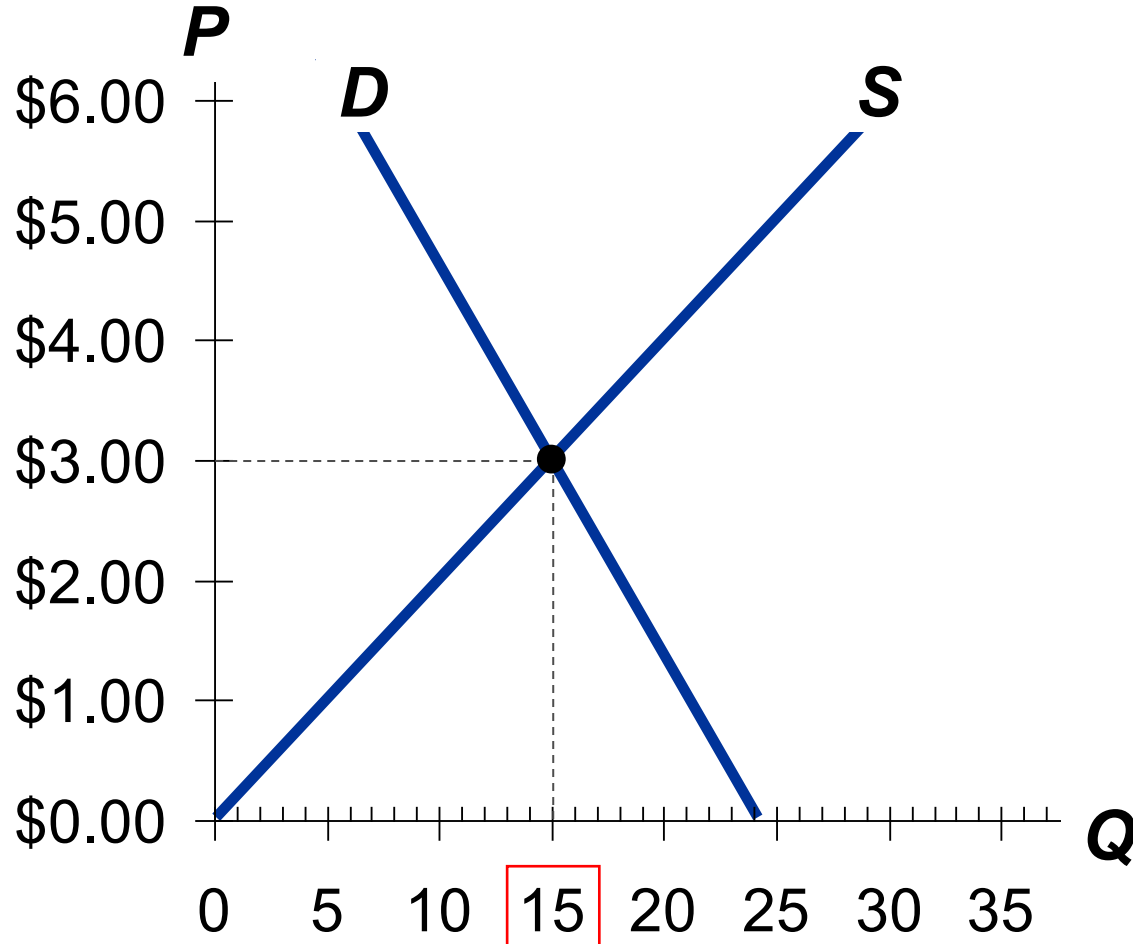
the price that equates quantity supplied with quantity demanded



| $P$ | $Q^D$ | $Q^S$ |
|-----|-------|-------|
| \$0 | 24    | 0     |
| 1   | 21    | 5     |
| 2   | 18    | 10    |
| 3   | 15    | 15    |
| 4   | 12    | 20    |
| 5   | 9     | 25    |
| 6   | 6     | 30    |

## Equilibrium quantity:

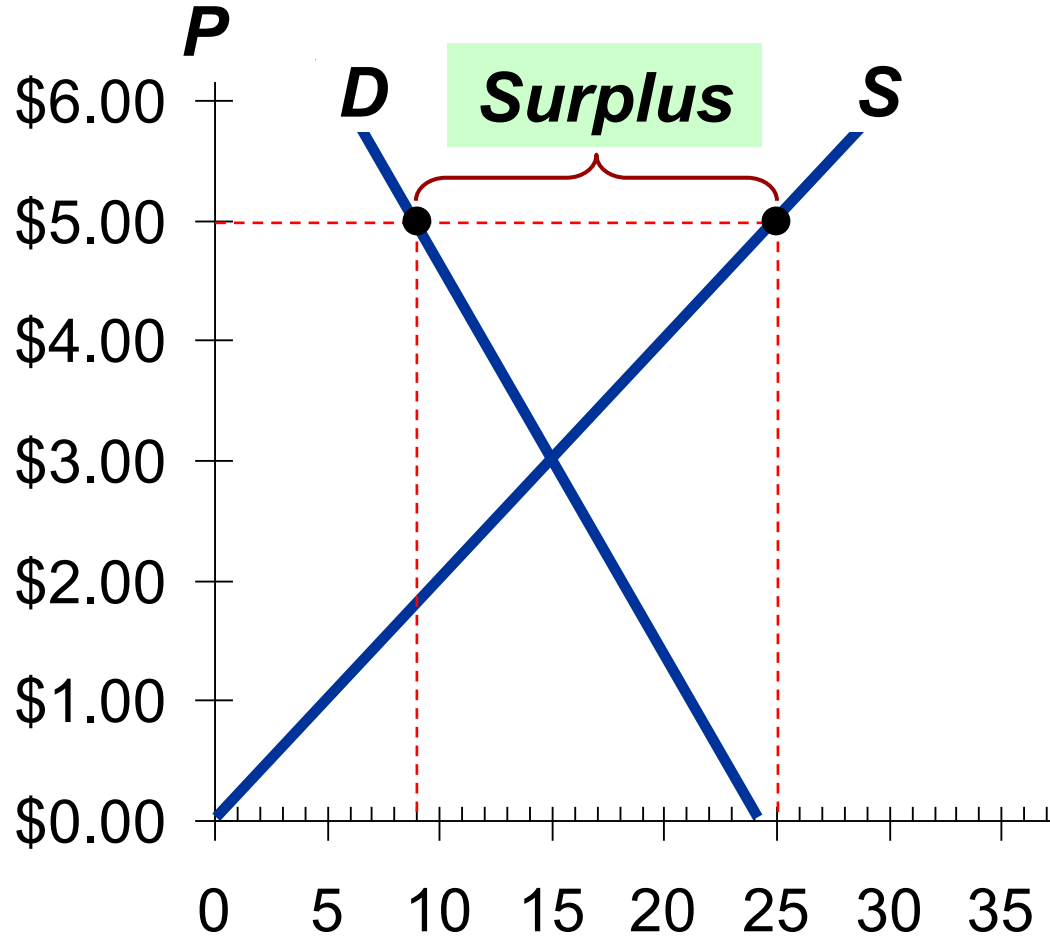
the quantity supplied and quantity demanded at the equilibrium price



| $P$ | $Q^D$ | $Q^S$ |
|-----|-------|-------|
| \$0 | 24    | 0     |
| 1   | 21    | 5     |
| 2   | 18    | 10    |
| 3   | 15    | 15    |
| 4   | 12    | 20    |
| 5   | 9     | 25    |
| 6   | 6     | 30    |

## Surplus (a.k.a. excess supply):

when quantity supplied is greater than quantity demanded



Example:

If  $P = \$5$ ,

then

$$Q^D = 9 \text{ lattes}$$

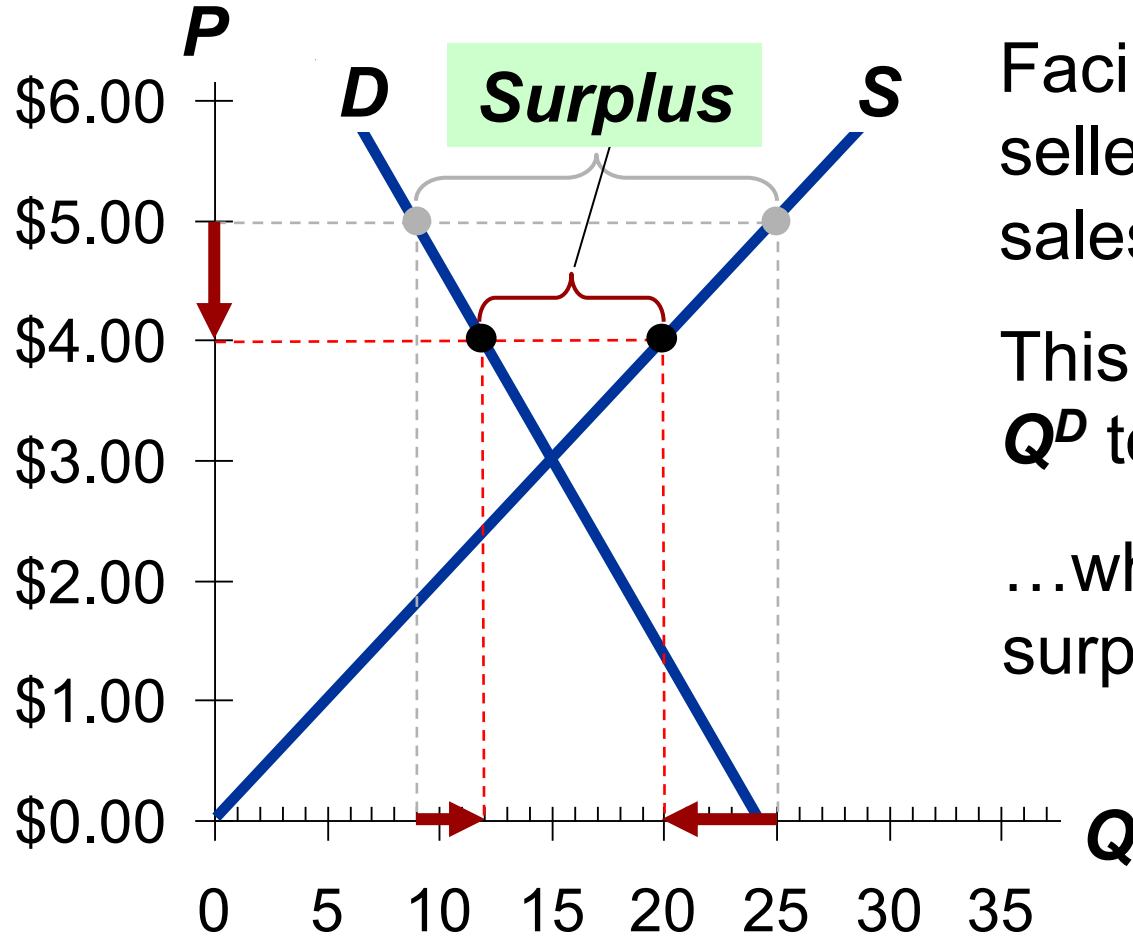
and

$$Q^S = 25 \text{ lattes}$$

resulting in a  
surplus of 16 lattes

## Surplus (a.k.a. excess supply):

when quantity supplied is greater than quantity demanded



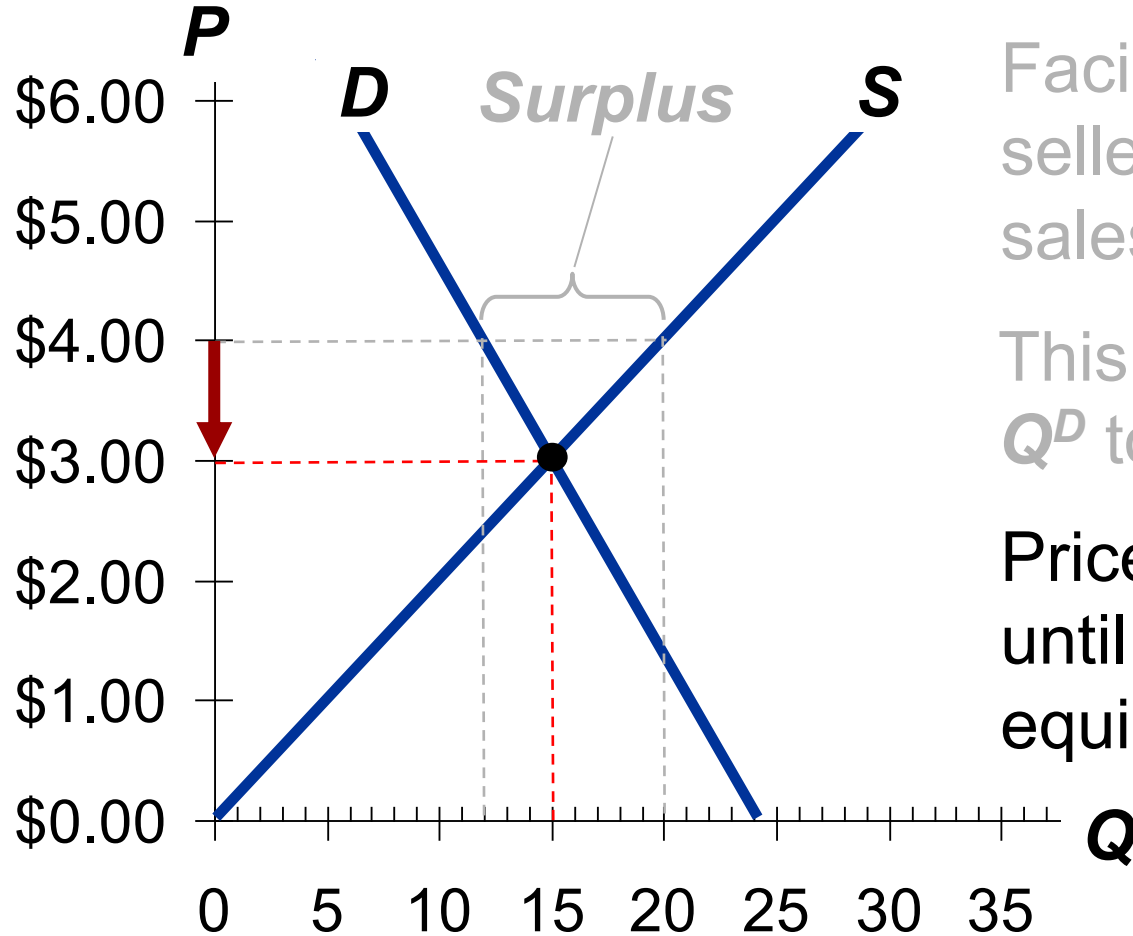
Facing a surplus, sellers try to increase sales by cutting price.

This causes  $Q^D$  to rise and  $Q^S$  to fall...

...which reduces the surplus.

## Surplus (a.k.a. excess supply):

when quantity supplied is greater than quantity demanded



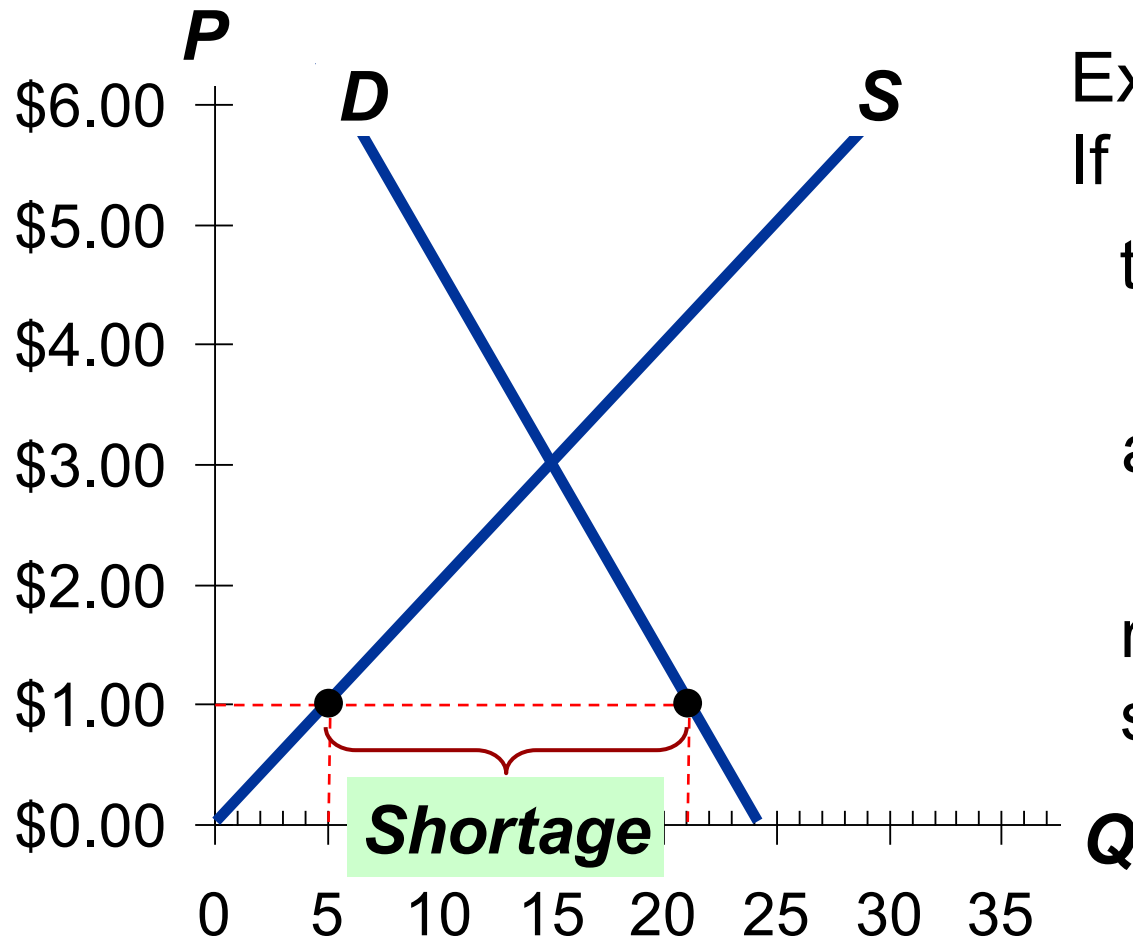
Facing a surplus, sellers try to increase sales by cutting price.

This causes  $Q^D$  to rise and  $Q^S$  to fall.

Prices continue to fall until market reaches equilibrium.

## Shortage (a.k.a. excess demand):

when quantity demanded is greater than quantity supplied



Example:

If  $P = \$1$ ,

then

$Q^D = 21$  lattes

and

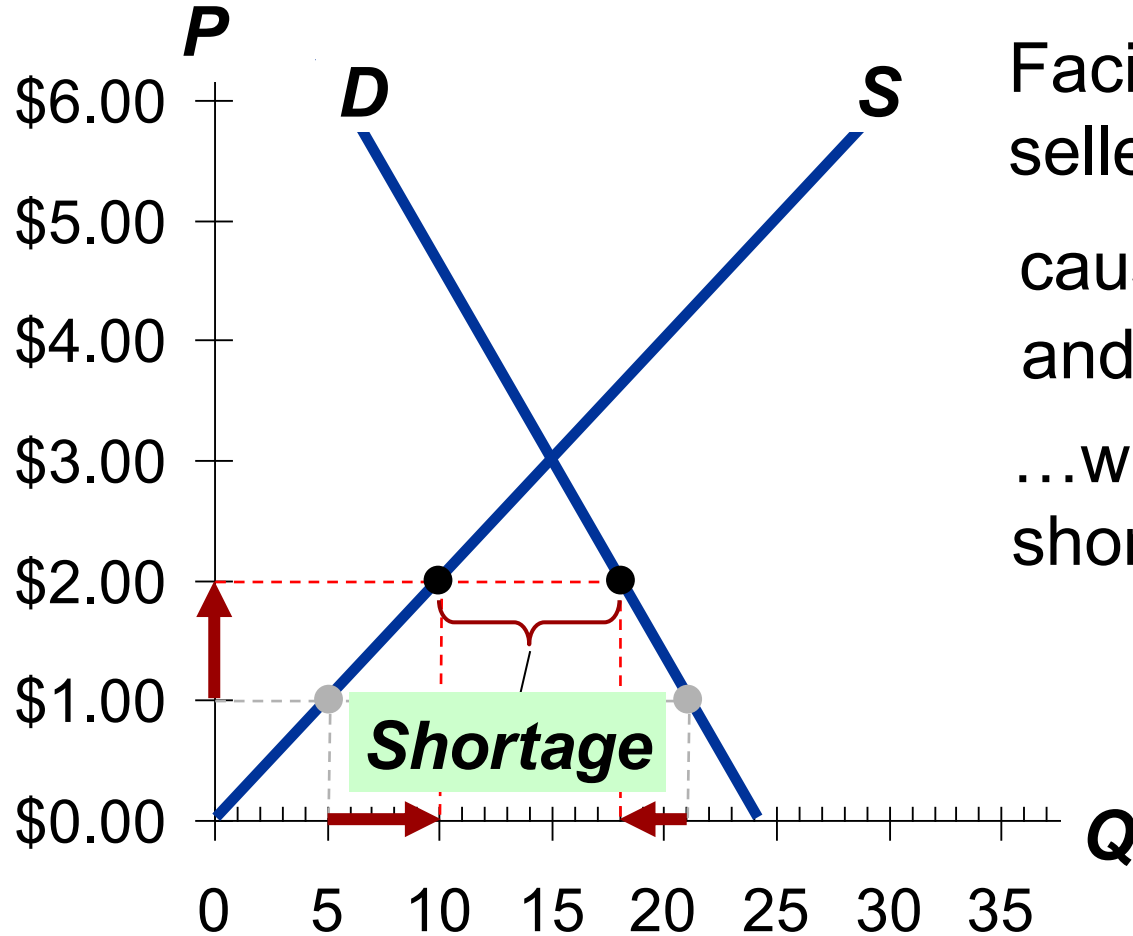
$Q^S = 5$  lattes

resulting in a

shortage of 16 lattes

## Shortage (a.k.a. excess demand):

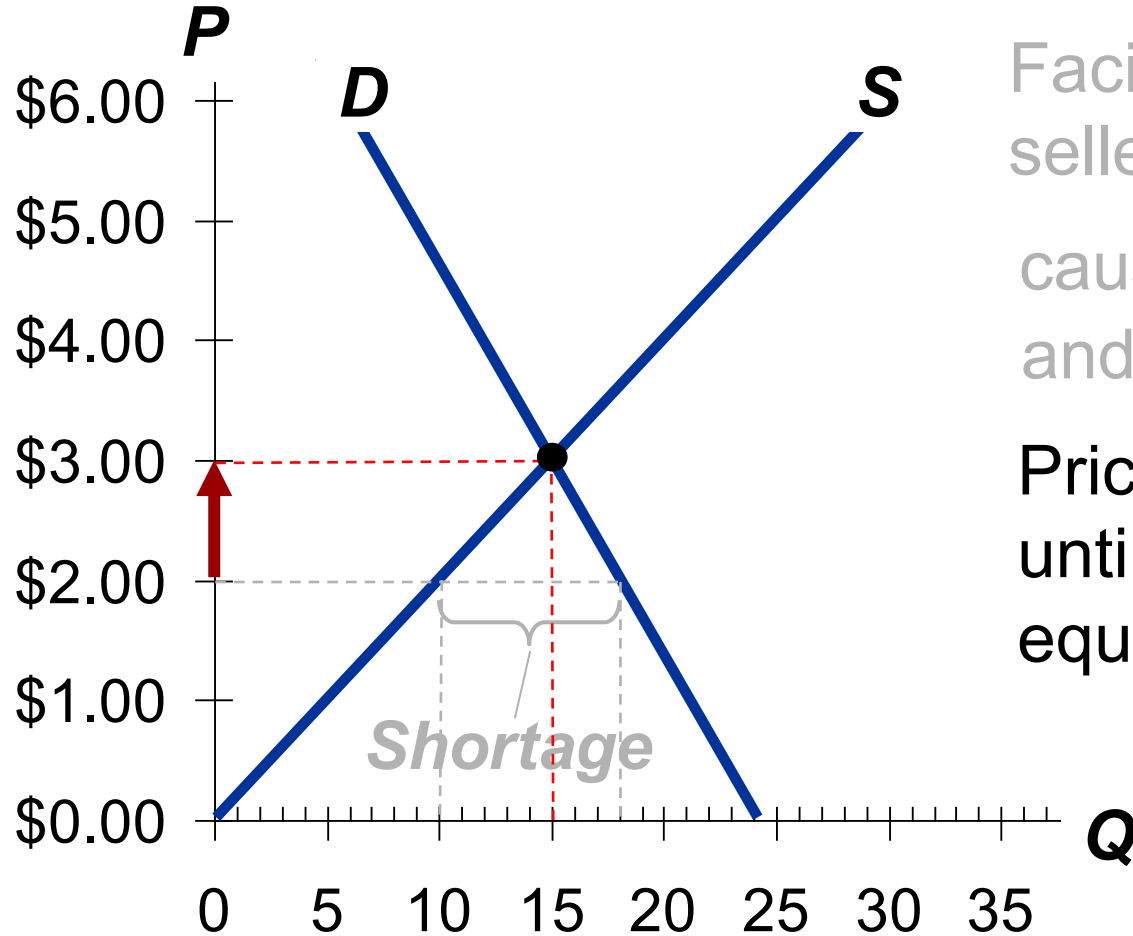
when quantity demanded is greater than quantity supplied



Facing a shortage, sellers raise the price, causing  $Q^D$  to fall and  $Q^S$  to rise, ...which reduces the shortage.

## Shortage (a.k.a. excess demand):

when quantity demanded is greater than quantity supplied



Facing a shortage, sellers raise the price, causing  $Q^D$  to fall and  $Q^S$  to rise.

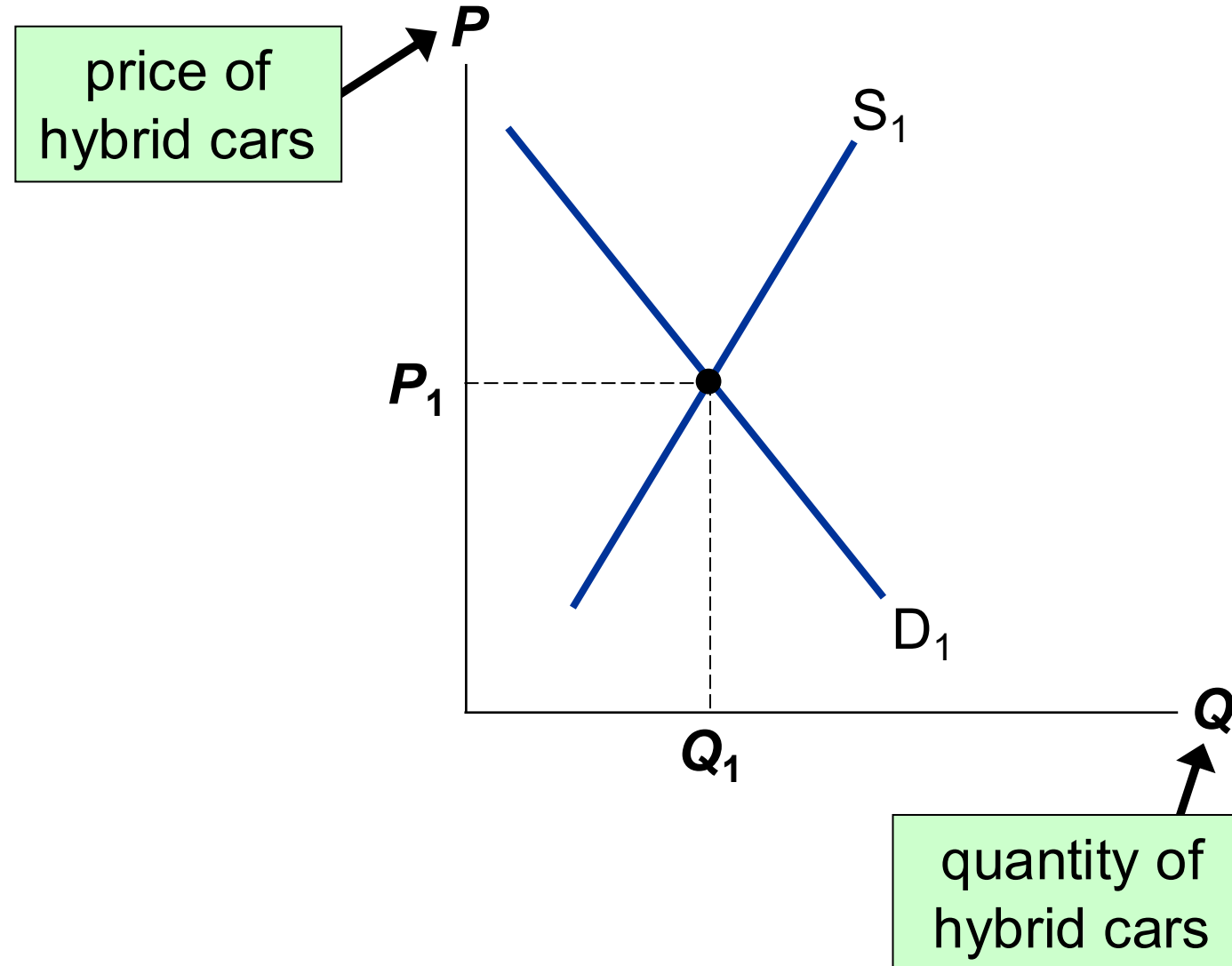
Prices continue to rise until market reaches equilibrium.

## *Three Steps to Analyzing Changes in Equilibrium*

To determine the effects of any event,

1. Decide whether event shifts ***S*** curve, ***D*** curve, or both.
2. Decide in which direction curve shifts.
3. Use supply—demand diagram to see how the shift changes eq'm ***P*** and ***Q***.

# EXAMPLE: The Market for Hybrid Cars



# EXAMPLE 1: A Shift in Demand

**EVENT TO BE  
ANALYZED:**

Increase in price of fuel.

**STEP 1:**

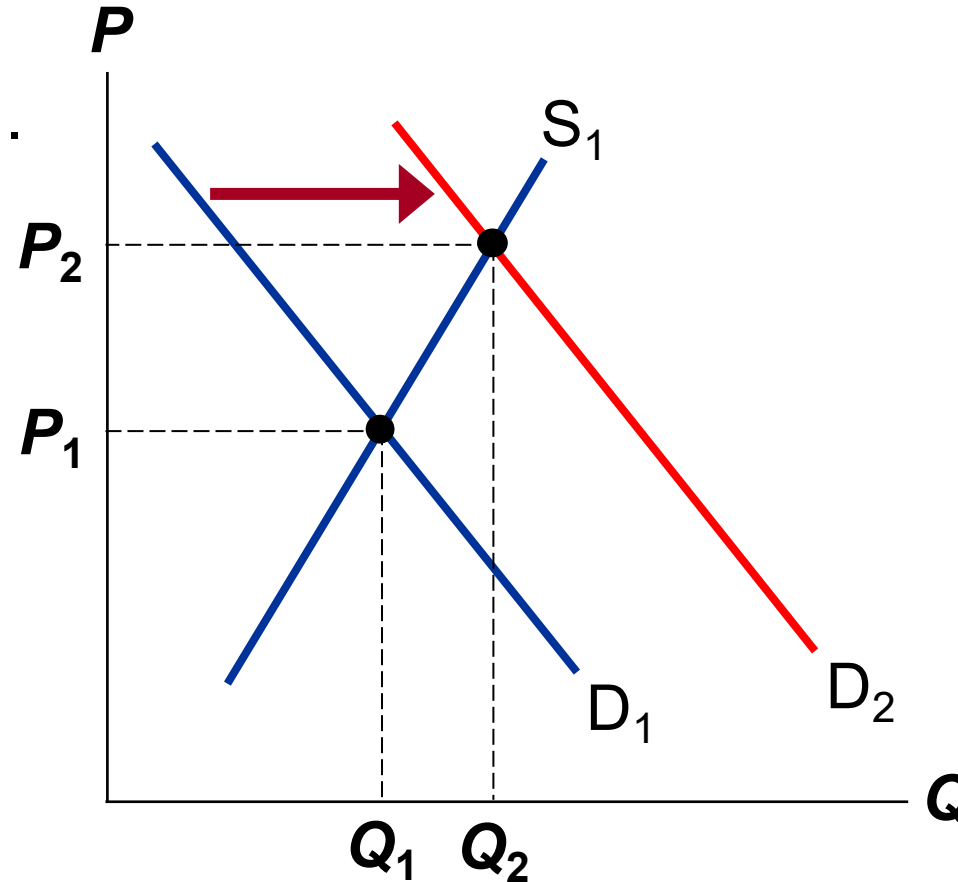
*D* curve shifts

**STEP 2:**

*D* shifts right

**STEP 3:**

The shift causes an increase in price and quantity of hybrid cars.

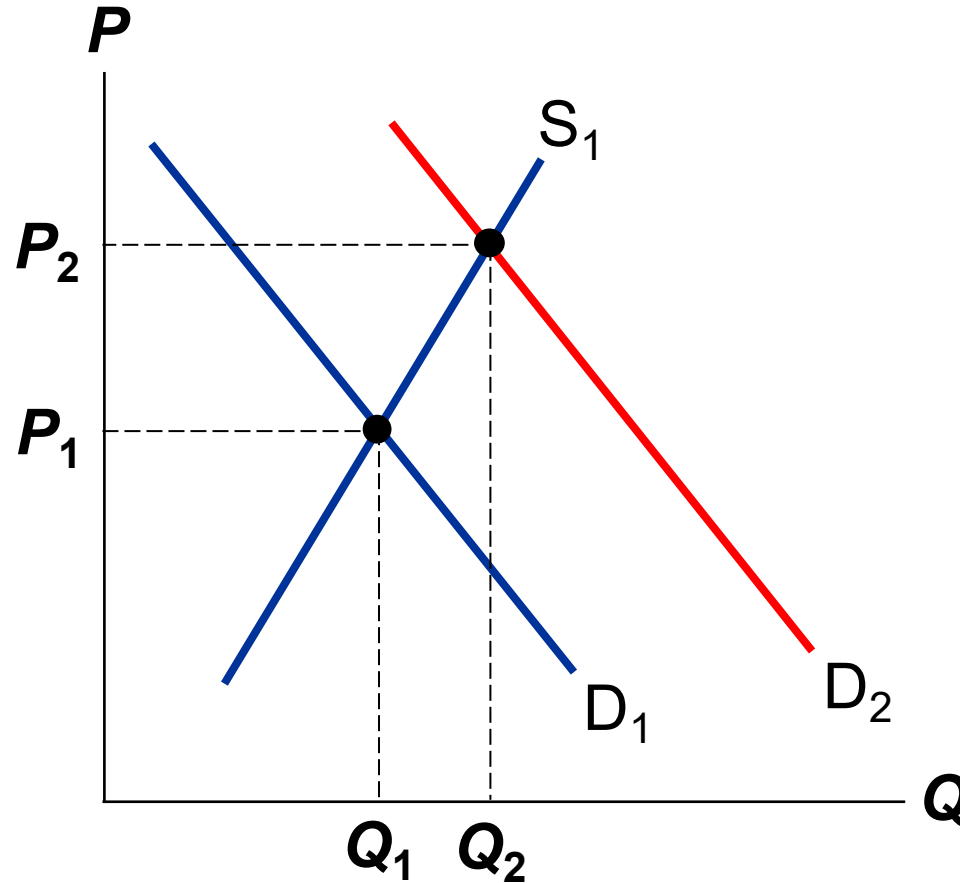


## EXAMPLE 1: A Shift in Demand

Notice:

When  $P$  rises, producers supply a larger quantity of hybrids, even though the  $S$  curve has not shifted.

***Always be careful to distinguish b/w a shift in a curve and a movement along the curve.***



## *Terms for Shift vs. Movement Along Curve*

### **Change in supply:**

a shift in the  $S$  curve occurs when a non-price determinant of supply changes (like technology or costs)

### **Change in the quantity supplied:**

a movement along a fixed  $S$  curve occurs when  $P$  changes

### **Change in demand:**

a shift in the  $D$  curve occurs when a non-price determinant of demand changes (like income or # of buyers)

### **Change in the quantity demanded:**

a movement along a fixed  $D$  curve occurs when  $P$  changes

## EXAMPLE 2: A Shift in Supply

**EVENT:** New technology reduces cost of producing hybrid cars.

**STEP 1:**

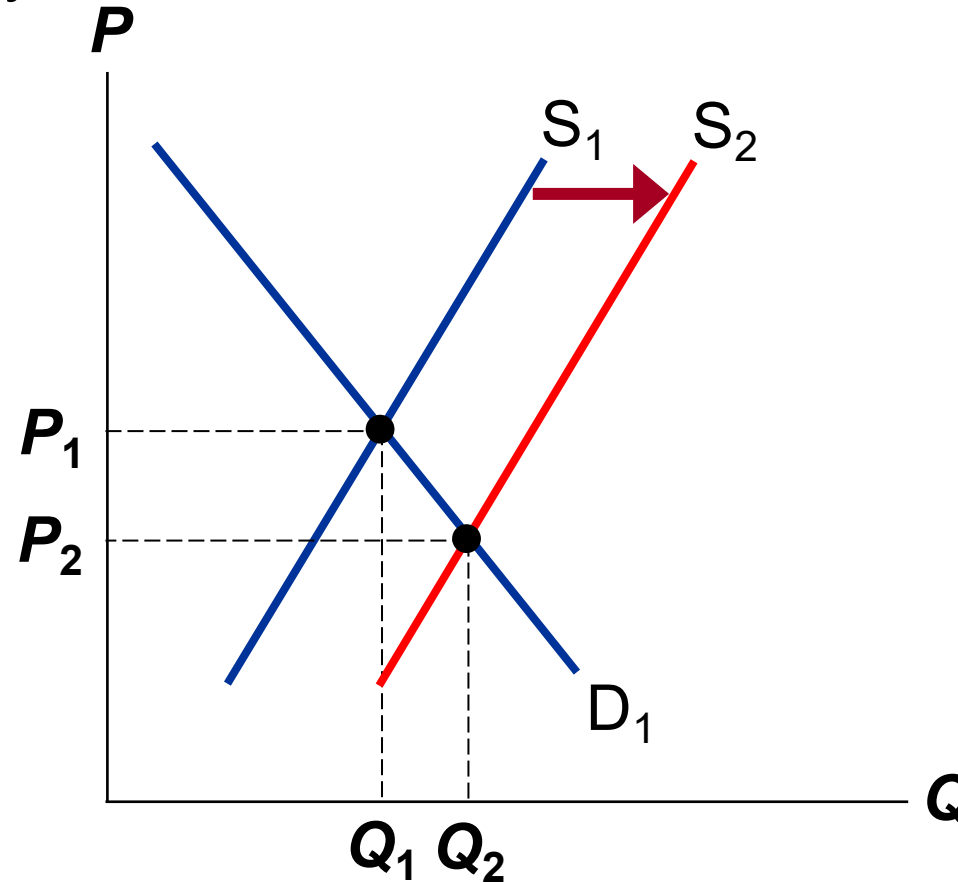
**S** curve shifts

**STEP 2:**

**S** shifts right

**STEP 3:**

The shift causes price to fall and quantity to rise.



## EXAMPLE 3: A Shift in Both Supply and Demand

### EVENTS:

Price of fuel rises AND  
new technology reduces  
production costs

### STEP 1:

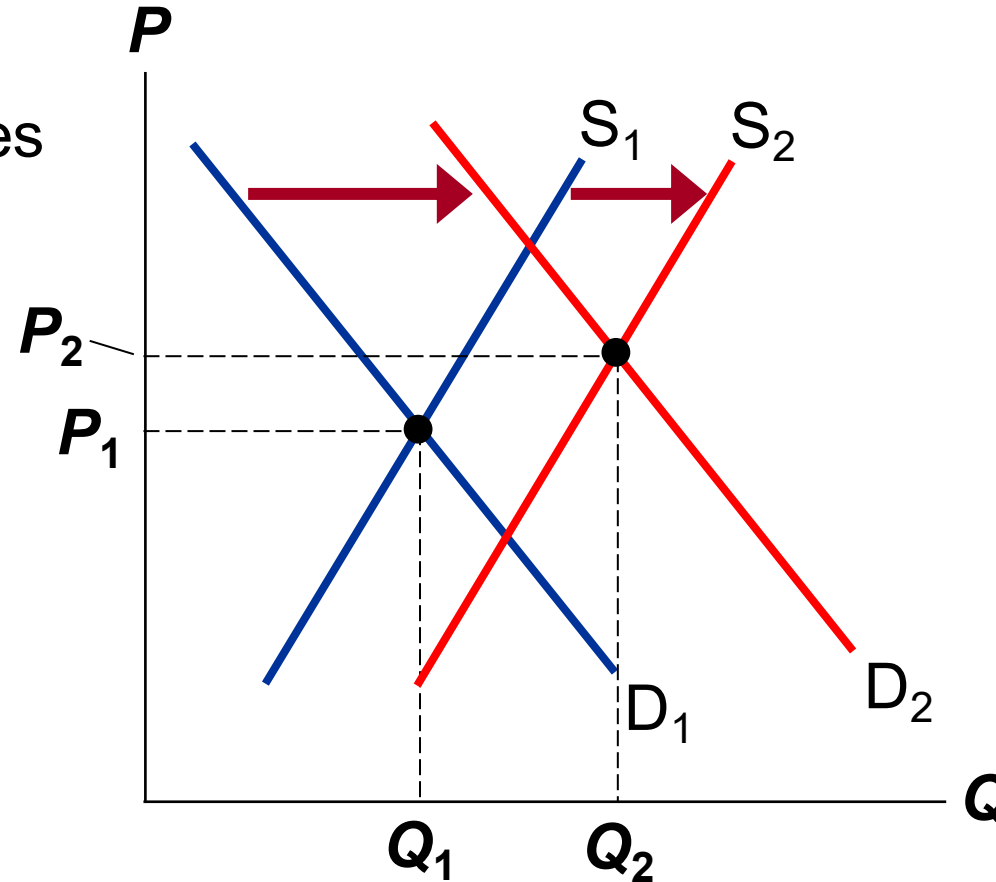
Both curves shift.

### STEP 2:

Both shift to the right.

### STEP 3:

$Q$  rises, but effect  
on  $P$  is ambiguous:  
If demand increases more  
than supply,  $P$  rises.



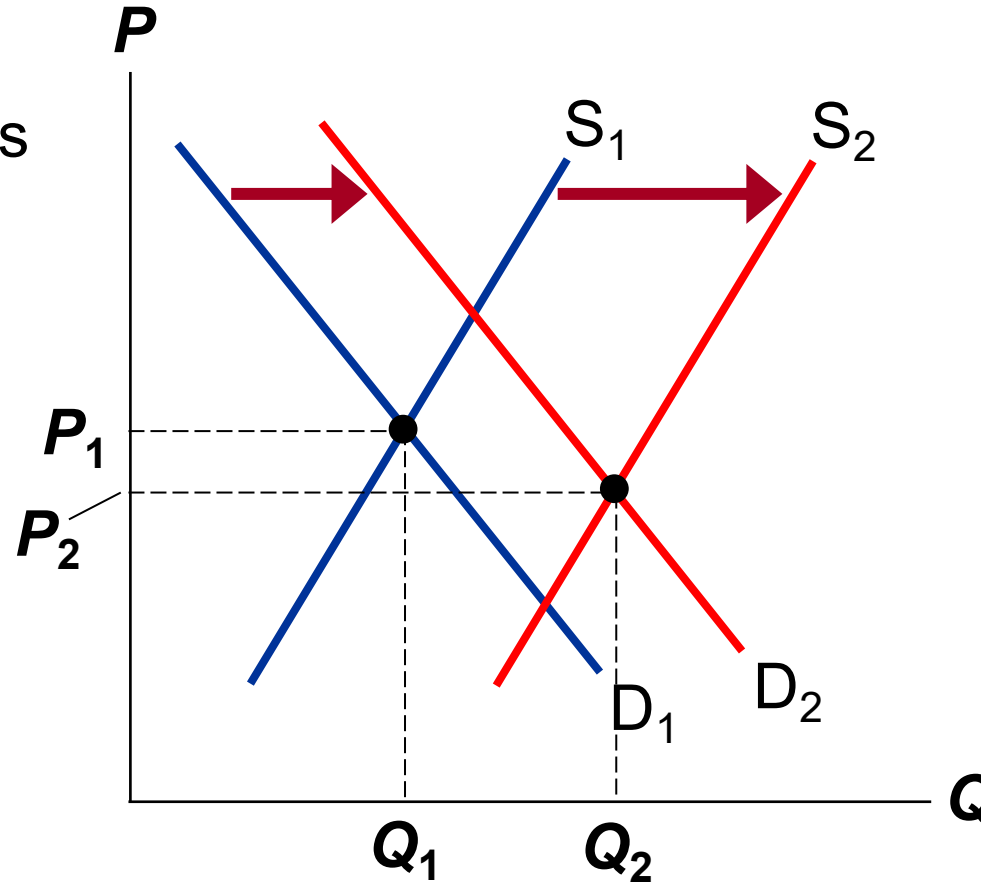
## EXAMPLE 3: A Shift in Both Supply and Demand

### EVENTS:

price of fuel rises AND  
new technology reduces  
production costs

### STEP 3, cont.

But if supply  
increases more  
than demand,  
 $P$  falls.



## Shifts in supply and demand

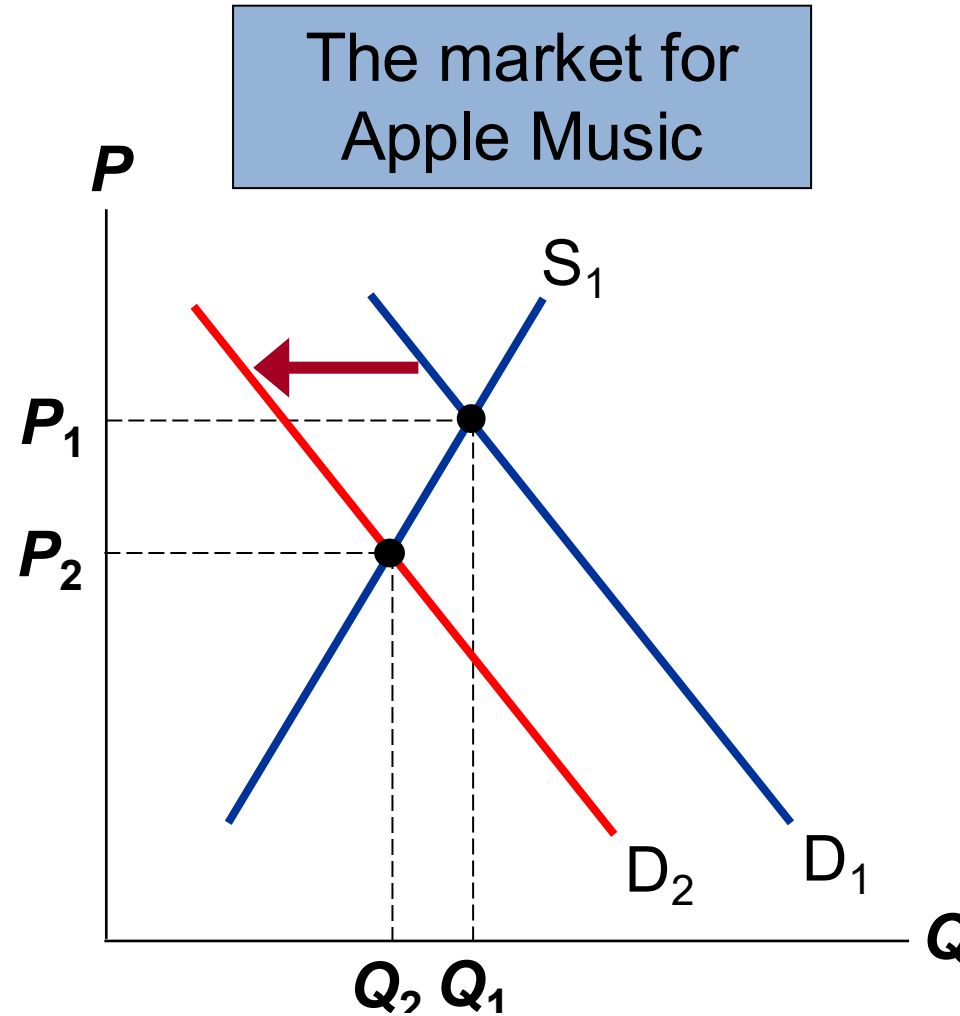
**Use the three-step method to analyze the effects of each event on the equilibrium price and quantity of Apple Music.**

- Event A: A fall in the price of Spotify Premium
- Event B: Sellers of Apple Music negotiate a reduction in the royalties they must pay for each song they sell.
- Event C: Events A and B both occur.

## A. Fall in price of Spotify Premium

### STEPS

1. ***D*** curve shifts
2. ***D*** shifts left
3. ***P*** and ***Q*** both fall.

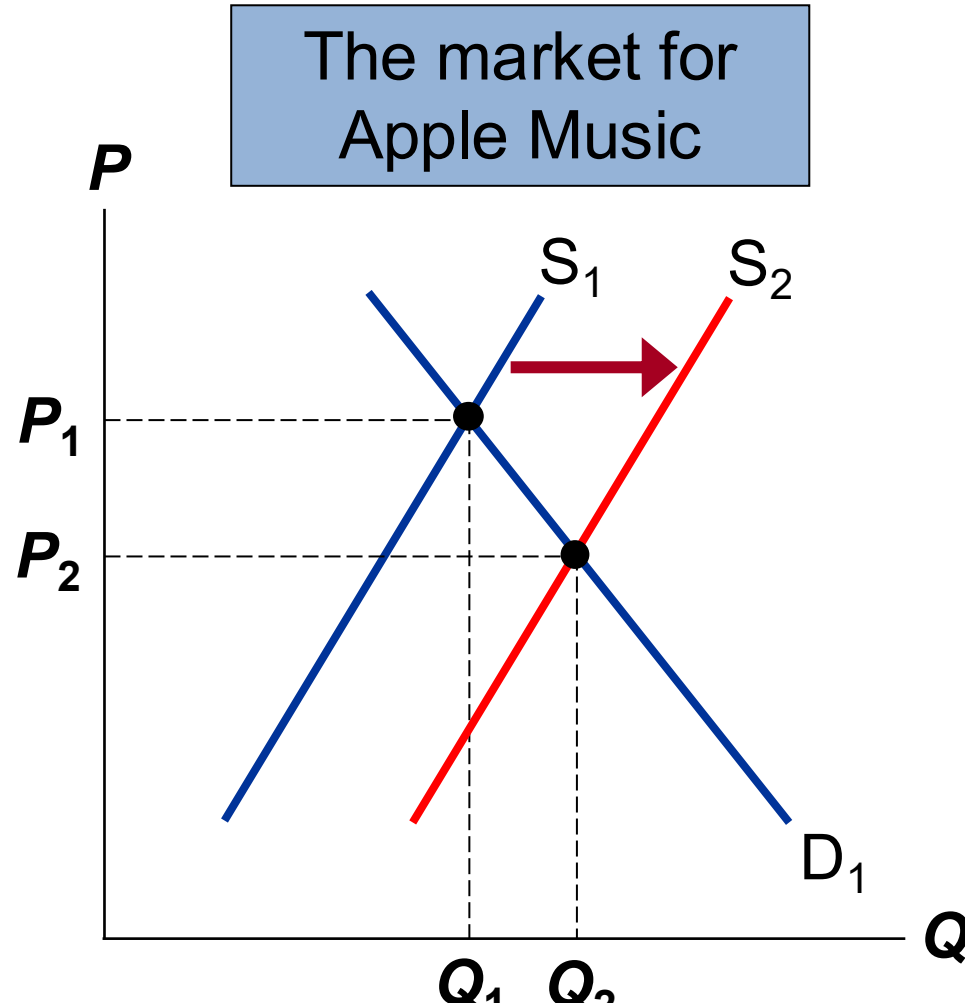


## B. Fall in cost of royalties

### STEPS

1. **S** curve shifts
2. **S** shifts right
3. **P** falls,  
**Q** rises.

(Royalties are part of sellers' costs)



## C. Fall in price of Spotify Premium and fall in cost of royalties

### STEPS

1. Both curves shift (see parts A & B).
2. ***D*** shifts left, ***S*** shifts right.
3. ***P*** unambiguously falls.

Effect on ***Q*** is ambiguous:

The fall in demand reduces ***Q***,  
the increase in supply increases ***Q***.

**CONCLUSION:**  
**How Prices**  
**Allocate**  
**Resources**

One of the Principles from Chapter 1:

*Markets are usually a good way  
to organize economic activity.*

In market economies, prices adjust to balance supply and demand. These equilibrium prices are the signals that guide economic decisions and thereby allocate scarce resources.

## **SUMMARY**

**A competitive market has many buyers and sellers, each of whom has little or no influence on the market price.**

**Economists use the supply and demand model to analyze competitive markets.**

## SUMMARY

The **downward-sloping demand curve** reflects the law of demand, which states that the quantity buyers demand of a good depends negatively on the good's price.

Besides price, demand depends on buyers' *incomes, tastes, expectations, the prices of substitutes and complements, and number of buyers*. **If one of these factors changes, the D curve shifts.**

The **upward-sloping supply curve** reflects the Law of Supply, which states that the quantity sellers supply depends positively on the good's price.

Other determinants of supply *include input prices, technology, expectations, and the # of sellers*. **Changes in these factors shift the S curve.**

## SUMMARY

The intersection of S and D curves determines the market equilibrium. **At the equilibrium price, quantity supplied equals quantity demanded.**

*If the market price is above equilibrium, a surplus results, which causes the price to fall.*

*If the market price is below equilibrium, a shortage results, causing the price to rise.*

## SUMMARY

**We can use the supply-demand diagram to analyze the effects of any event on a market:**

- **Determine whether the event shifts one or both curves.**
- **Determine the direction of the shifts.**
- **Compare the new equilibrium to the initial one.**

**In market economies, prices are the signals that guide economic decisions and allocate scarce resources.**